Amylase and pH: Continuous-Sampling Investigation

LAUNCH · Science · 40 minutes

I can follow the centre-approved Pearson method to investigate how pH affects amylase activity, recording safe continuous-sampling evidence accurately.

Prepare before pupils arrive

Setup: Technician and teacher must prepare, label and check all solutions and equipment under the current centre-approved Pearson method and risk assessment. Allocate roles, prepare a dry/paper parity route and decide the exact endpoint rule before pupils enter.

Safety: This deck is not a method sheet, COSHH assessment or certification record. Follow the latest centre/Pearson instructions. Control iodine, heating, solution concentrations, contamination, spills and disposal; never use pupil-generated method details as authorisation.

Equipment: Only the apparatus and prepared solutions listed in the centre's current Pearson-aligned CP 1.10 method: typically labelled amylase, starch and buffer solutions, iodine spotting tile setup, clean transfer equipment, reaction vessels, timer and authorised temperature-control equipment.

Safety: Teacher/technician control applies throughout. Wear required eye protection; iodine is hazardous and stains; avoid skin/eye contact and inhalation, never taste or mouth-pipette, keep solutions labelled, follow heating and spill procedures, and use only current COSHH/local risk controls. Stop immediately if the authorised setup differs from the deck summary.

Fallback: Take a safe non-contact role—timer, interval caller, observer or exact recorder—or analyse teacher-provided raw observations. The teacher decides whether any practical evidence meets course requirements.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Edexcel GCSE Biology 1BI0 Foundation · Paper 1 · Topic 1 · specification point 1.10 investigation of pH and enzyme activity.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 3 min	State the independent and dependent variables. Name two variables that must stay constant.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today’s answer.
Starter · 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask ‘What makes you think that?’ and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	continuous sampling — test small samples over time endpoint — the stopping observation validity — tests the intended question Teacher checks goggles, labelled solutions, tile layout, clean transfer equipment, timing roles and the authorised temperature-control arrangement before permission to begin. Prepare the specified small iodine volume in each labelled tile well before mixing amylase, buffer and starch in the authorised order.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Keep roles clear: one trained pupil samples, one calls the interval, one records, and one checks the tile map; rotate roles only when the teacher says it is safe. Use separate labelled transfer equipment as directed to reduce cross-contamination.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 5 min	Before or during the physical practical, classify each situation as continue, stop/check, or record-and-repeat. Choose one live-lab situation.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 12 min	Carry out the assigned part of the centre-approved amylase-pH investigation and preserve a complete raw evidence record. Record: A dated raw table with pH, each sampled colour, first observed endpoint time, anomalies/uncertainty and the pupil's own method note.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	State the endpoint for one trial and explain why a shorter time indicates greater amylase activity. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Give one next move; the pupil edits or re-attempts on the existing work.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: 90 s, the first sampled time at which starch was not detected, with interval uncertainty acknowledged.

Correct. Record the first sampled time meeting the agreed endpoint; the true change occurred somewhere within the sampling interval, so 75 s is not known exactly.

Misconception repair

Repaired. Preserve the raw observation, flag uncertainty and repeat under the approved method if appropriate.

Guided checkpoint

Correct. Record the first sampled time meeting the agreed endpoint; the true change occurred somewhere within the sampling interval, so 75 s is not known exactly. Use only times actually observed. Do not invent an exact time between samples.

Ready tile

CONTINUE AS AUTHORISED. Continue: the checked setup is ready for the authorised next method step. Look for evidence that method, safety and roles have all been checked.

Iodine spill

STOP AND CHECK ADULT. Stop and alert the adult. Follow the local spill/first-aid procedure; do not improvise cleanup. A chemical spill requires the trained adult and local procedure.

Unlabelled buffer

STOP AND CHECK ADULT. Stop: an unlabelled solution is unsafe and cannot provide valid pH evidence. Both chemical identity and IV value must be known before use.

Missed interval

RECORD, THEN REPEAT IF AUTHORISED. Record the missed sample honestly and repeat only if the authorised method, time and materials allow. Do not invent the missing colour or quietly change the sampling time.

Borderline colour

RECORD, THEN REPEAT IF AUTHORISED. Record the ambiguity and repeat or moderate the judgement using the authorised protocol. Uncertain evidence should be preserved and checked, not forced into a preferred result.

Clean fixed sample

CONTINUE AS AUTHORISED. Continue: the sampling action follows the checked method. This situation has no safety or evidence fault to repair.

Temperature outside range

STOP AND CHECK ADULT. Stop and check the adult. The method condition and safety setup need correction before valid continuation. Temperature is both a controlled variable and part of the authorised apparatus setup.

Model limitation

The iodine colour is a threshold judgement, not a continuous measure of starch concentration. Sampling intervals limit time precision, and the classroom method does not show molecular collisions or active-site shape directly.

Independent evidence

A dated raw table with pH, each sampled colour, first observed endpoint time, anomalies/uncertainty and the pupil's own method note. The page itself makes no CP-completion claim.

Supported: Complete one clearly labelled reaction timeline with an adult-authorised role; circle the first orange-brown endpoint and state whether starch is detected. Use the colour key, pre-labelled time row and exact scribing/pointing route. Keep the same scientific endpoint.

Standard: Run or record assigned pH trials, identify endpoint times and annotate one control-variable or uncertainty check. Use the prepared raw table. Write what was observed before writing what it means.

Stretch: Preserve full sampling evidence, estimate the maximum endpoint-time uncertainty from the interval and justify whether a repeat is needed. Separate an ambiguous endpoint, missed sample and anomalous result; do not imply greater precision than the method provides.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre’s authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner’s practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_L_W8L2_Amylase_pH_Core_Practical_Explore.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.